C++

程式語言（二）

Introduction to Programming (II)

Classes & Objects

Joseph Chuang-Chieh Lin

Dept. CSE, NTOU

Platform/IDE

- Dev-C++

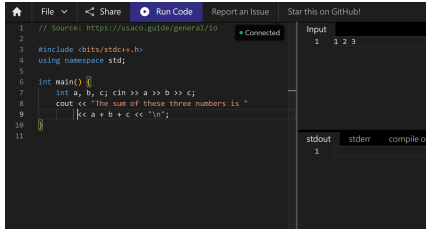
https://www.pngegg.com/en/search?q=Dev-C

- Codeblocks

https://icons8.com/icons/set/code-blocks

- OnlineGDB (https://www.onlinegdb.com/)

- Real-Time Collaborative Online IDE (https://ide.usaco.guide/)



Textbooks (We focusing on C++11)

- ***Learn C++ Programming by Refactoring* (由重構學習 C++ 程式設計). Pang-Feng Liu (劉邦鋒). NTU Press. 2023.**
- ***C++ Primer. 5th Edition.* Stanley B. Lippman, Josée Lajoie, Barbara E. Moo. 2019.**
- *Effective C++*. Scott Meyers. O'Reilly. 2016.
- *Thinking in C++. Vol. 1: Introducing to Standard C++*. 2nd Edition. Bruce Eckel. Prentice Hall PTR. 2000.

Useful Resources

- Tutorialspoint
  - https://www.tutorialspoint.com/cplusplus/index.htm
  - Online C++ Compiler
- Programiz
  - https://www.programiz.com/cpp-programming
- LEARN C++
  - https://www.learncpp.com/
- MIT OpenCourseWare - Introduction to C++
  - https://ocw.mit.edu/courses/6-096-introduction-to-c-january-iap-2011/pages/lecture-notes/
- Learning C++ Programming
  - https://www.programiz.com/cpp-programming
- GeeksforGeeks
  - https://www.geeksforgeeks.org/c-plus-plus/

Classes & Objects

Class

- A blueprint/prototype/sketch for the **object**.
- To design a “car”, you need:
  - Wheels
  - Engines
  - A steering wheel
  - Windows
  - Lights
  - …
  -

Class

- To design a “school”, you need:
  - Buildings
    - Windows
    - Chairs
    - A Blackboard
    - ...
  - Teachers
  - Students
  - Walls
  - Staffs

Class

- We define our own data structures by defining a **class**.
- A class defines a **type** along with a collection of **operations** that are related to that type.
- A primary focus of the design of C++ codes is to make it possible to define **class types** that *behave as naturally as the built-in types*.

Create a class (by an example)

```
class Room {
    public:
        double length;
        double breadth;
        double height;

        double calculateArea() {
            return length * breadth;
        }

        double calculateVolume() {
            return length * breadth * height;
        }
};
```

data members
資料成員

member functions
成員函式

Create a class (by an example)

```
class Room {
    public:
        double length;
        double breadth;
        double height;

        double calculateArea(){
            return length * breadth;
        }

        double calculateVolume(){
            return length * breadth * height;
        }

};
```

```
int main() {
    // create objects of Room
    Room r1, r2;

    //assign values to data members
    r1.length = 42.5;
    r1.breadth = 30.8;
    r1.height = 19.2;

    //calculate the area
    cout << "area: ";
    cout << r1.calculateArea();
    cout << endl;
}
```

Create a class (by an example)

different instances!

```
class Room {
    public:
        double length;
        double breadth;
        double height;

        double calculateArea(){
            return length * breadth;
        }

        double calculateVolume(){
            return length * breadth * height;
        }

};
```

```
int main() {
    // create objects of Room
    Room r1, r2;

    //assign values to data members
    r1.length = 42.5;
    r1.breadth = 30.8;
    r1.height = 19.2;

    //calculate the area
    cout << "area: ";
    cout << r1.calculateArea();
    cout << endl;
}
```

The keyword “public” and “private”

- **public**: the members in the class can be assessed anywhere from the program.
- **private**: the members can only be assessed from within the class (i.e., member functions).
  - Information hiding

Example of using **private**

```
class Room {
    private:
        double length;
        double breadth;
        double height;
    public:
        void initData(double len, double brth, double hgt) {
            length = len;
            breadth = brth;
            height = hgt;
        }
        double calculateArea(){
            return length * breadth;
        }
        double calculateVolume(){
            return length * breadth * height;
        }
};
```

Example of using **private**

```
class Room {
    private:
        double length;
        double breadth;
        double height;
    public:
        void initData(double len, double brth,
            length = len;
            breadth = brth;
            height = hgt;
        }
        double calculateArea(){
            return length * breadth;
        }
        double calculateVolume(){
            return length * breadth * height;
        }
};
```

```
int main() {
    // create objects of Room
    Room r1;

    //initial the Room object
    r1.initData(42.5, 30.8, 19.2);

    //calculate the area
    cout << "area: ";
    cout << r1.calculateArea();
    cout << endl;

    cout << "volume: ";
    cout << r1.calculateVolume();
    cout << endl;
}
```

Example: Animal

```
class Animals {
   public:
      void sound() {
         cout << "Animals' sound."
            << endl;
      }
};
```

```
int main (){
   Animals dog, cat;
   dog.sound();
   cat.sound();
   return 0;
}
```

A Boring Example

```
class Homework {
    public:
        void DO();
};
void Homework::DO() {
    // need to "#include <utility>" for swap
    string s1;
    getline(cin, s1);
    auto len {s1.length()};
    string s2 {s1};
    for (unsigned int i=0; i <len/2; i++)
        swap(s2.at(i), s2.at(len-i-1));
    cout << s2 << endl;
}
```

```
int main (){
    Homework h1;
    h1.DO();
    return 0;
}
```

https://onlinegdb.com/K3lwpNE8T

Learn from a well-designed class in a header file.

The Bookstore transactions

- We shall learn how to “use” a class.
- `Sales_item.h` [link to download]

```
#include <iostream>
#include "Sales_item.h"
int main()
{
    Sales_item book;
    // read ISBN, number of copies sold, and sales price
    std::cin >> book;
    // write ISBN, number of copies sold, total revenue, and average price
    std::cout << book << std::endl;
    return 0;
}
```

The Bookstore transactions

- Sample input of previous code:

```
0-201-70353-X 4 24.99
```

- The sample output:

```
0-201-70353-X 4 99.96 24.99
```

The Bookstore transactions - Adding `Sales_items`

```
#include <iostream>
#include "Sales_item.h"
int main()
{
    Sales_item item1, item2;
    std::cin >> item1 >> item2;
    // read a pair of    transactions
    std::cout << item1 + item2 << std::endl;
    // print their sum
    return 0;
}
```

Sample input of previous code:

```
0-201-78345-X 3 20.00
0-201-78345-X 2 25.00
```

The sample output:

```
0-201-70353-X 5 110 22
```

The Bookstore transactions
- Example of Member functions/methods

```
#include <iostream>
#include "Sales_item.h"
int main()
{
    Sales_item item1, item2;
    std::cin >> item1 >> item2;
    // first check that item1 and item2 represent the same book
    if (item1.isbn() == item2.isbn()) {
        std::cout << item1 + item2 << std::endl;
        return 0; // indicate success
    } else {
        std::cerr << "Data must refer to same ISBN"
                  << std::endl;
        return -1; // indicate failure
    }
}
```

The Bookstore Program

```
#include <iostream>
#include "Sales_item.h"
int main()
{
     Sales_item total; // variable to hold data for the next transaction
     // read the first transaction and ensure that there are data to process
     if (std::cin >> total) {
          Sales_item trans; // variable to hold the running sum
          // read and process the remaining transactions
          while (std::cin >> trans) {
               // if we're still processing the same book
               if (total.isbn() == trans.isbn())
                    total += trans; // update the running total
               else {
                    // print results for the previous book
                    std::cout << total << std::endl;
                    total = trans; // total now refers to the next book
               }
          }
          std::cout << total << std::endl; // print the last transaction
     } else {
          // no input! warn the user
          std::cerr << "No data?!" << std::endl;
          return -1; // indicate failure
     }
     return 0;
}
```

Global Scope/Function Scope /Block Scope

Refer to: https://en.cppreference.com/w/cpp/language/scope
https://www.geeksforgeeks.org/scope-of-variables-in-c/

```
#include <iostream>
using namespace std;

//global
int global = 5;

int main() {
    //local variable within a function scope
    int global = 2;
    cout << global << endl;

    return 0;
}
```

Global variables can be assessed from ANY part of the program.

- Usually declared outside all of the functions or blocks.

Global Scope/Function Scope /Block Scope

Refer to: https://en.cppreference.com/w/cpp/language/scope
https://www.geeksforgeeks.org/scope-of-variables-in-c/

```
#include <iostream>
using namespace std;

int number = 1;

int main ()
{
  cout << "Global number: " << number << endl;
  int number = 2;
  cout << "Local number: " << number << endl;
  {
    int number = 3;
    cout << "Block number: " << number << endl;
  }
  return 0;
}
```

```
Global number: 1
Local number: 2
Block number: 3
```

Variables defined in a block “{ }” can be seen inside that block!

Namespace Scope

- Purpose of using namespaces: [reference link]
  - Organize code into logical groups.
    - Prevent name collisions (namespace pollution).
  - Suitable for teamwork in case collisions happen.

- Usage:

  `using namespace::name;`

- Note :

  Don't use `using` in a header file.

  (potential unexpected name collisions)

```
#include <iostream>
using std:cin;

int main()
{
      int i;
      cin >> i;
      cout << i; // error!;
      std:cout << i; // OK!
      return 0;
}
```

Namespace Examples

https://www.geeksforgeeks.org/namespace-in-c/

```
#include <iostream>
using namespace std;

namespace first
{
    int val = 500;
}

int val = 100;

int main()
{
    int val = 200;

    // These variables can be accessed from outside the namespace using the scope operator ::
    cout << first::val << endl;

    return 0;
}
```

Namespace in the same file

```
#include <iostream>
using namespace std;

namespace NTOU {
    class Student{
    public:
        string name; int age;
        double height; double weight;
        string department;
    };
    class Professor{
    public:
        string name; int age;
        double height; double weight;
        string department;
    };
}
```

```
int main() {
    NTOU::Student s1;
    s1.age = 20;
    s1.name = "Betty";
    return 0;
}
```

Namespace in different files

- Suppose that we have
  - `main.cpp`
  - Files for NTOU:
    - `ntou.h`
    - `ntou.cpp`
  - Files for CMU
    - `cmu.h`
    - `cmu.cpp`

Codes on OnlineGDB

Class assignment:
Modify the code (line 19) to have the following output:

```
Betty is playing in CMU!
Betty is studying hard in CMU!
```

Class Scope

- Every class defines its own new scope.
- Outside the class scope, ordinary **data and function members** may be accessed only through an **object**, a **reference**, or a **pointer** using a **member access operator**

  . or **->**

- We access **type members** from the class using the **scope operator : :**
- In either case, the name that follows the operator must be a member of the associated class.

Class Scope (Example)

(click to the code in my GitHub page)

```
class rectangle {
public:
     typedef int unit;
     void area();
     void set(unit wd, unit ht);
private:
     unit width;
     unit height;
};
```

```
void rectangle::set(unit wd, unit ht)
{
     width = wd;
     height = ht;
}
```

```
void rectangle::area() {
  cout << "The area: " << width * height << endl;
}
```

```
int main()
{
     rectangle obj, *obj2; //creating object of rectangle class
     rectangle::unit x, y;
     cin >> x;
     cin >> y;
     obj.set(x, y);
     obj2 = &obj;
     obj.area();
     obj2->area();
     return 0;
}
```

Class Scope (Example)

(click to the code in my GitHub page)

https://onlinegdb.com/NsLIANoyo

```
class rectangle {
public:
     typedef int unit;
     void area();
     void set(unit wd, unit ht);
private:
     unit width;
     unit height;
};
```

```
void rectangle::set(unit wd, unit ht)
{
     width = wd;
     height = ht;
}
```

define the member function outside a class

```
void rectangle::area() {
  cout << "The area: " << width * height << endl;
}
```

```
int main()
{
     rectangle obj; //creating object of rectangle class
     rectangle::unit x, y;
     cin >> x;
     cin >> y;
     obj.set(x, y);
     rectangle &obj2 = obj; // this usage of & is special in C++
     obj.area();
     obj2.area();
     return 0;
}
```

Illustrating Example:

Structure & Constant Member Function

```
#include <iostream>
#include <string>
using namespace std;

struct Person {
// default: everything is public
    string name;
    int age;
    double salary;
};

// structure variable as an argument here
void display(const Person&);

int main() {
    Person p {"Joseph", 45, 10000000};
    display(p);
    return 0;
}
void display(const Person& p) {
    cout << "Name: " << p.first_name << endl;
    cout << "Age: " << p.age << endl;
    cout << "Salary: " << p.salary;
}
```

Let’s play around this program.

```
#include <iostream>
#include <string>
using namespace std;

struct Person {
// default: everything is public
    string name;
    int age;
    double salary;
    // structure variable as an argument here
    void display(const Person&) const;
};

int main() {
    Person p {"Joseph", 45, 10000000};
    p.display(p);
    return 0;
}

void Person::display(const Person& p) const {
    cout << "Name: " << p.name << endl;
    cout << "Age: " << p.age << endl;
    cout << "Salary: " << p.salary;
}
```

Constant member function of a class

https://onlinegdb.com/mXYp4x-kN

```
#include <iostream>
#include <string>
using namespace std;

struct Person {
// default: everything is public
    string name;
    int age;
    double salary;
    // structure variable as an argument here
    void display() const;
};

int main() {
    Person p {"Joseph", 45, 10000000};
    p.display();
    return 0;
}

void Person::display() const {
    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "Salary: " << salary;
}
```

Another way (more concise)

```
#include <iostream>
#include <string>
using namespace std;

struct Person {
// default: everything is public
    string name;
    int age;
    double salary;
    // structure variable as an argument here
    void display(double salary) const;
};
int main() {
    Person p {"Joseph", 45, 10000000};
    p.display(100);
    cout << endl;
    cout << p.salary << endl;
    return 0;
}
void Person::display(double salary) const {
    this->salary += salary;
    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "Salary: " << salary;
}
```

Using this pointer...

error: assignment of member ‘Person::salary’ in read-only object

```
#include <iostream>
#include <string>
using namespace std;

struct Person {
// default: everything is public
    string name;
    int age;
    double salary;
    // structure variable as an argument here
    void display(doubld salary) const;
};
int main() {
    Person p {"Joseph", 45, 10000000};
    p.display(100);
    cout << endl;
    cout << p.salary << endl;
    return 0;
}
void Person::display(double salary) const {
    this->salary += salary;
    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "Salary: " << salary;
}
```

Using this pointer…
(UPDATED VERSION)

```
Name: Joseph
Age: 45
Salary: 100
10100
```

https://onlinegdb.com/6hNFd9zrrA

Exercise - BOOK Specification

Define a class **`BOOK`** with the following specification

- Private members:
  - **`BOOK_NO`**: (int)
  - **`TITLE`**: 20 characters (string)
  - **`PRICE`**: float (price per copy)
  - **`TOTAL_COST(N)`** : float (a function calculating the total cost for `N` copies; `N` is passed as argument)

- Public member functions:
  - **`INPUT()`** : Function to read `BOOK_NO`, `TITLE`, and `PRICE`.
  - **`PURCHASE()`** : Function to ask the user to input the number of copies to be purchased. It invokes `TOTAL_COST()` and prints the total cost to be paid by the user.

Sample Input & Output

C:\CPP\book_purchase.exe

```
In INPUT():
Enter Book Title: 海大資工讚讚讚
Enter Book Number: 20240301
Enter price per copy: 999
In PURCHASE:
Enter number of copies to purchase: 10
Total cost: 9990

Process returned 0 (0x0)   execution time : 0.329 s
Press any key to continue.
```

The main function:

```
int main()
{
    BOOK obj;
    obj.INPUT();
    obj.PURCHASE();
    return 0;
}
```

Another Example

Define a class **`student`** with the following specification

- Private members:
  - **`studentID`**: (int)
  - **`name`**: 20 characters (string)
  - **`eng`**, **`math`**, **`phy`**: float
  - **`total`**: float (the sum of eng, math, and phy)
  - **`grades()`**: a function to calculate eng + math + phy with float return type.
- Public member functions:
  - **`Takedata()`**: Function to accept values for `studentID`, `name`, `eng`, `math` grades (float type) and invoke `grades()` to calculate `total`.
  - **`Showdata()`**: Function to display all the data members on the screen.

Sample Input & Output

```
C:\Users\josep\_programs\PL\students.exe
In Takedata():
Enter studentID: 693410001
Enter student name: 林金毛
Enter grades in English, Math, and Physics: 100 99 92
In Showdata():
StudentID: 693410001
Student Name: 林金毛
English: 100
Math: 99
Physics: 92
Total: 291
--------------------------------
Process exited after 30.34 seconds with return value 0
請按任意鍵繼續 . . .
```

The main function:

```
int main ()
{
    student obj ;
    obj.Takedata();
    obj.Showdata();
    return 0;
}
```

Another Exercise

Define a class **`student`** with the following specification

- Private members:
  - **`studentID`**: (int)
  - **`name`**: 20 characters (string)
  - **`eng`**, **`math`**, **`phy`**: float
  - **`avg()`**: a function to calculate the average grade of `eng`, `math`, and `phy` with float return type.
- Public member functions:
  - **`Takedata()`**: Function to accept values for `studentID`, `name`, `eng`, `math` grades and invoke `avg()` to calculate the average grade.
  - **`Showdata()`**: Function to display all the data members on the screen.
  - **`PassOrFail()`**: Function to display "**`pass`**" if the average grade >= 60 or "**`fail`**". otherwise.

Sample Input & Output

C:\Users\josep_Study_Programming Language\C++\hw_2.exe

```
In Takedata()
Enter studentID: 693410001
Enter student name: 張大勇
Enter grades in English, Math, and Physics: 100 90 80
In Showdata()
StudentID:693410001
Student Name:張大勇
English:100
Math:90
Physics:80
Pass


--------------------------------
Process exited after 19.95 seconds with return value 0
請按任意鍵繼續 . . .
```

The main function:

```
int main ()
{
    student obj ;
    obj.Takedata();
    obj.Showdata();
    obj.PassOrFail();
    return 0;
}
```

Glocal, and Static Variables

When a local variable has the same name as a global variable...

```
#include<iostream>
using namespace std;

int val;  // val is global

int main()
{
    int val = 10; // val is local here
    cout << "Value of global val is " << ::val;
    cout << "\nValue of local val is " << val;
    return 0;
}
```

```
Value of global val is 0
Value of local val is 10
```

To access a static variable in a class

```
using namespace std;

class Test {
    static int x;
public:
    static int y;

    void func(int x) {
        cout << "Value of static x is " << Test::x;

        cout << "\nValue of local x is " << x;
    }
};
```

```
// static members must be explicitly
// defined
int Test::x = 1;
int Test::y = 2;

int main() {
    Test obj;
    int x = 3;
    obj.func(x);

    cout << "\nTest::y = " << Test::y;
    return 0;
}
```

```
Value of static x is 1
Value of local x is 3
Test::y = 2
```

To access a static variable in a class

https://onlinegdb.com/5H2zEGyWk

```
using namespace std;

class Test {
    static int x;
public:
    static int y;

     void increase() {
          x++;
          y++;
     }
     void show() {
          cout << "x: " << x << endl;
          cout << "y: " << y << endl;
     }
};
```

```
// static members must be explicitly
// defined
int Test::x = 1;
int Test::y = 2;

int main() {
     Test obj1, obj2;
     obj1.increase();
     obj1.show();
     obj2.increase();
     obj2.show();
}
```

```
x: 2
y: 3
x: 3
y: 4
```

To access a static variable in a class

https://onlinegdb.com/5H2zEGyWk

```
using namespace std;

class Test {
    static int x;
public:
    static int y;

     void increase() {
          x++;
          y++;
     }
     void show() {
          cout << "x: " << x << endl;
          cout << "y: " << y << endl;
     }
};
```

```
// static members must be explicitly
// defined
int Test::x = 1;
int Test::y = 2;

int main() {
     Test obj1, obj2;
     obj1.increase();
     obj1.increase();
     obj2.show();
     return 0;
}
```

```
x: 3
y: 4
```

Global vs. Static

Feature	Global Variable	Static Global	Static Local (in Function)
Lifetime	Entire program	Entire program	Entire program
Scope	Whole Project (via `extern`)	Only that **file**	Only that **function**
Initialization	Once, at start	Once, at start	Once, first time called
Best Use Case	Constants/Config	File-specific settings	Tracking state in a function

Global

- Declared **outside** of any function or class.
- **Scope**: It is visible to the entire file and *can even be accessed by other files in your project* using the `extern` keyword.
- **Lifetime**: It’s created when the *program starts* and destroyed when it ends.
- **Risk**: Because anyone can change it, global variables can lead to "spaghetti code" where it's *hard to track* who modified the value.

Static

- **Static Global:**
  - **Visibility**: Other files cannot see or use this variable, even with `extern`. It prevents naming conflicts if another file has a variable with the same name.
  - **Usage:** when some data should be **private** to one .cpp file.
- **Static Local:**
  - **Scope**: It is local to that function (you can't access it outside).
  - **Lifetime**: Unlike regular local variables tha ‘‘die’’ when the function ends, a static variable remembers its value between function calls.
  - **Usage:** when a function needs to remember something from previous calls.

Example of “Static Global”

https://onlinegdb.com/ED0GJT7su3

file1.cpp

```cpp
#include <iostream>
using namespace std;

// Counter is only visible in file1.cpp
static int Counter = 0; // file private

void increaseCounter() {
    fileCounter++;
    cout << "file1.cpp: Counter = "
        << Counter << endl;
}
```

main.cpp

```cpp
void increaseCounter();
void tryAccess();

int main() {
    increaseCounter();
    increaseCounter();
    tryAccess();
    return 0;
}
```

file2.cpp

```cpp
#include <iostream>
using namespace std;

// extern int Counter;
// ERROR if you try to access it from file2.cpp here

void tryAccess() {
    cout << "file2.cpp cannot directly access Counter" << endl;
}
```

Example of “Static Local”

```
#include <iostream>
using namespace std;

int getNextID() {
    static int nextID = 1000;
    return nextID++;
}

int main() {
    cout << getNextID() << endl; // 1000
    cout << getNextID() << endl; // 1001
    cout << getNextID() << endl; // 1002
    return 0;
}
```

```
#include <iostream>
using namespace std;

void setupSystem() {
    static bool initialized = false;

    if (!initialized) {
        cout << "Doing initialization...\n";
        initialized = true;
    }
    cout << "System is running.\n";
}

int main() {
    setupSystem();//Doing initialization...
    setupSystem();//System is running.
    SetupSystem();//System is running.
    return 0;
}
```

Another Class Example: `Circle`

```
#include <iostream>
using namespace std;

class Circle
{
    private :
        double radius; // data members
    public:
        void setRadius(double r);
        double getArea(); //member functions
};

void Circle::setRadius(double r)
{
    radius = r;
}

double Circle::getArea()
{
    return 3.14 * radius * radius;
}
```

```
int main()
{
    Circle c1; //an object of class circle
    c1.setRadius(2.5); //call member function
                       // to initialize radius
    cout << c1.getArea(); //display the area
    return 0;
}
```

Discussions & Questions